

Halo communication in the mEVP sea-ice solver of the FESOM2 GPU port

N. Koldunov, 6 August 2026 — summary of measurements

Three ways of exchanging the halo less often

mEVP advances the ice momentum in 120 sub-cycles per model time step, and the ice-velocity halo is at present exchanged after each of them. We implemented and measured three alternatives.

Delayed exchange — exchange every K -th sub-cycle and accept a halo up to $K - 1$ sub-cycles old. This is the largest saving of the three, but it leaves the sub-domain boundaries visible in the ice field. *Wide halo* (Danilov) — exchange every K -th sub-cycle, and recompute a K -deep ring of ghost cells in between so that no value is stale; the result reproduces the every-sub-cycle trajectory bit for bit. *Divergence form* (Danilov) — carry $\nabla \cdot \sigma$ at nodes rather than σ at elements; it agrees with $\nabla \cdot \sigma$ to 10^{-15} relative, and its effect on run time differs between the two backends.

Cost, measured against the model step and against the ice

Each number is given twice. Sea ice is 5 to 13% of a CPU model step, so an ice-only change is divided by about ten before it reaches the step, and the share of the ice cost (ice, ice dynamics and ice advection, computation together with time spent waiting for communication) is the measure appropriate to a sea-ice scheme.

		GPU		CPU	
		ice cost	model step	ice cost	model step
delayed exchange	$K=2$	−32 to −39%	−8.3 to −9.3%	−2 to −18%	−0.1 to −2.9%
delayed exchange	$K=4$	−37 to −58%	−8.5 to −14%	−5 to −25%	−0.3 to −3.6%
wide halo, exact	$K=2$	−15% [†]	+1.7 to −3.8%	never profitable	
wide halo, exact	$K=8$	−12 to −19%	−2 to −7%	never profitable	
divergence form alone	—	+2 to +17%	0 to +1.6%	−2 to −8.5%	−0.5 to 0%

GPU ranges span the four meshes, each at the node count where it still gains from more GPUs: CORE2 on 1 node (4 GPUs), fArc on 4, DARS on 8, NG5 on 16. CPU ranges span 128 to 2048 cores on CORE2 and fArc. [†] one point only (DARS, 8 nodes); the wide halo at $K=2$ is a cost on fArc at 4 nodes and a saving at 16, so it needs the larger K .

On CPU the saving grows with core count (−5% of the ice at 128 cores, −25% at 2048); on GPU it falls as each GPU holds more of the mesh (−58% at 32 000 surface nodes per GPU, −37% at 460 000). The divergence form removes 2 to 8.5% of the ice cost from 512 cores upward, and is invisible in the model step only because of the denominator. On GPU it costs, because the sub-cycle there is limited by how often work is dispatched to the device rather than by arithmetic or memory traffic, so removing element arrays removes work that was not being paid for. Danilov’s ~20% for the divergence form with its double halo corresponds to our −19.1% of the ice cost for the same combination (fArc, 16 GPU nodes).

Why the delayed exchange is not usable

Under the delayed exchange the stale values lie on the one-cell boundary of each sub-domain, and the decomposition is fixed for the run, so the same nodes are perturbed at every step. On fArc with 16 GPUs over 60 days, $|\Delta \text{ ice thickness}|$ averaged on those boundaries is 2.5 times its interior value at $K=4$ and 1.7 times at $K=2$, against 0.85 for two identical runs and 0.87 for the exact wide halo. The difference is concentrated there from the first day and appears on maps as a network following the sub-domain edges. A polygonal structure in the ice field, in the shape of the domain decomposition, is not acceptable in a sea-ice model, and that ends the matter — the scheme’s accuracy in the integral diagnostics is beside the point. Section 6.6 and Figure 4 of the full note give the profiles, the map and the controls.

Recommendation

For production we would use the exact wide halo. It is bit-for-bit identical to the standard scheme, so it requires neither an accuracy argument nor the check above, and it removes 12 to 19% of the ice cost on GPU at $K=8$. On CPU it is not profitable on any mesh we tried.

We do not regard the delayed exchange as usable, although it saves about three times as much: it places the domain decomposition in the ice field.

The divergence form is worth having on CPU at large core counts and is a cost on GPU. It is also what makes the wide halo cheapest to carry, since everything sub-cycled is then a node quantity.